

Project Design Phase
Proposed Solution Template

Date	09 July 2026
Team ID	SWTID-2026-2791
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Farmers face challenges such as unpredictable weather, inefficient irrigation, excessive fertilizer usage, crop diseases, low productivity, and lack of real-time decision support. These issues increase farming costs and reduce crop yield and profitability.
2.	Idea / Solution description	The Smart Agricultural Optimization Engine is an AI and IoT-based intelligent farming system that monitors soil moisture, weather conditions, crop health, and environmental data in real time. It provides personalized recommendations for irrigation, fertilizer application, crop selection, disease detection, and yield prediction through a user-friendly mobile application, enabling farmers to make data-driven decisions.
3.	Novelty / Uniqueness	• Integrates AI, Machine Learning, and IoT into a single platform. • Provides personalized recommendations instead of generic advice. • Real-time monitoring of soil, weather, and crop conditions. • Early disease detection using image processing and AI. • Predictive analytics for crop yield and resource optimization. • Supports sustainable and precision agriculture.
4.	Social Impact / Customer Satisfaction	• Improves farmers' income by increasing productivity. • Reduces water and fertilizer wastage. • Promotes sustainable farming practices. • Helps farmers make informed decisions. • Enhances food security through higher crop yields. • Increases customer satisfaction by providing accurate and timely recommendations.

5.	Business Model (Revenue Model)	• Subscription-based premium services for farmers. • Sale or rental of IoT sensor kits. • Partnerships with agricultural organizations and government agencies. • Premium analytics for agribusiness companies. • Commission through fertilizer, seed, and equipment partners.
6.	Scalability of the Solution	The solution can be expanded across different regions and crop types by integrating local weather and soil datasets. It supports multiple languages and can be deployed as a cloud-based platform, making it suitable for millions of farmers. Future enhancements include drone integration, satellite imagery, blockchain-based crop traceability, and smart marketplace connectivity.